7.2 BASIC PRINCIPLE OF OPERATION DC GENERATOR

According to Faraday's law of electromagnetic induction, 'when a conductor is rotated in a magnetic field, to cut the magnetic lines of flux, dynamically induced emf is produced in the conductor. This emf causes a current to flow if the conductor circuit is closed'. The direction of the current is found by Fleming's Right-hand rule.

Hence, the basic requirements for the dynamically induced emf to exits are:

1) A steady magnetic field
2) A conductor or coils
3) Relative motion between the magnetic field and the conductors.

To understand the principle of EMF generation, consider the following Fig 7.1.

7.2.1 Elementary DC Generator

Consider a single-turn rectangular copper coil ABCD rotating clockwise in a magnetic field provided by N and S magnets or electromagnets are shown in fig 7.1. The two ends of the coil are joined to two split-rings 'a' and 'b' which are insulated from each other and from central shaft. Two collecting brushes (of carbon or copper) press against the split-rings. Their function is to collect the current produced in the coil and to carry it to the external load resistance R. The rotating coils are called as 'armature ' and the magnets as "field magnets" or "poles". The coil is rotated in clockwise direction at a uniform magnetic field.

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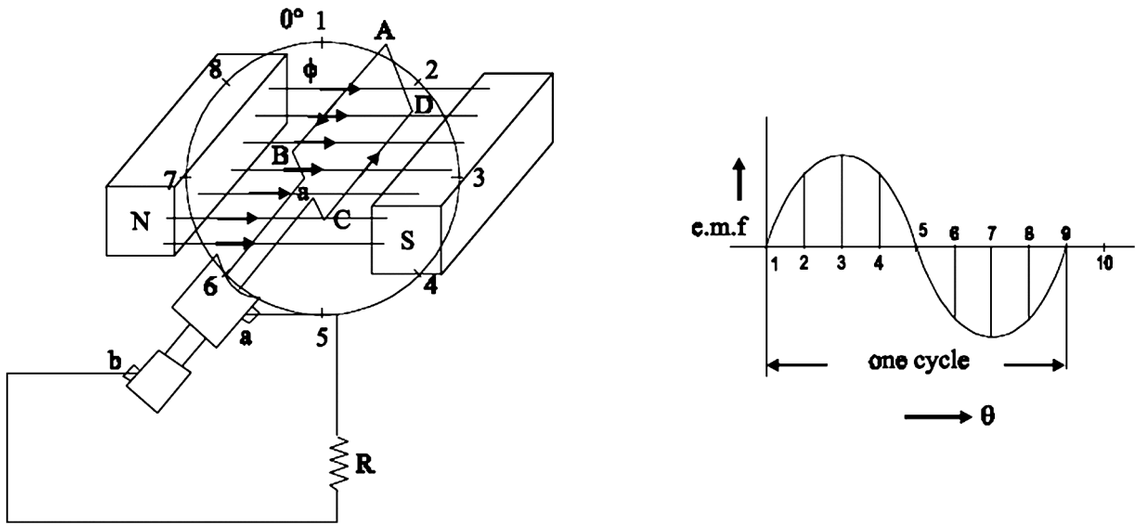


Fig 7.1 *Principle of EMF Generation*

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